VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

Reservation System with availability check

A8514- Database Management Systems Laboratory

Submitted by

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VARDHAMAN COLLEGE OF ENGINEERING

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This is to certify that the Course End Project titled "Reservation System with **availability check"** is carried out by **CHEEMALA HARINI REDDY (24881A0508)** towards **A8514,RYAKKAM LIKITHA(24881A0553),C ESWARI SAI PRAVALLIKA(25885A0501) Database Management System Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during the Academic year 2025-26.

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Abstract

The Hotel Booking Schema with Stored Procedure for Availability Check project aims to design and implement a structured database system for managing hotel reservations efficiently. The system is built using a relational database model that organizes data into tables such as customers, rooms, bookings, and payments, ensuring proper data management and integrity.

A major feature of this project is the use of a stored procedure to check room availability. This procedure takes user inputs like check-in and check-out dates and verifies whether rooms are available, reducing manual effort and improving the speed of operations. It also helps maintain consistency by performing logic directly within the database.

Overall, the project demonstrates how database design and procedural SQL can be combined to create an effective hotel management system. It highlights the importance of query optimization, reduced redundancy, and efficient data handling, making it suitable for real-world applications.

Keywords: DBMS, Hotel Booking System, Relational Schema, Stored Procedure, Availability Check, SQL, Database Design, Data Integrity, Query Optimization, Reservation Management

1 Introduction

In today's digital era, efficient data management plays a crucial role in the smooth functioning of businesses, especially in the hospitality industry. Hotels require reliable systems to manage customer information, room details, and booking records in an organized and error-free manner. Traditional manual methods of maintaining reservation records are time-consuming, prone to errors, and inefficient when handling large volumes of data.

This project focuses on designing a Hotel Booking Schema that efficiently organizes data into structured tables such as customers, rooms, bookings, and payments. Proper relationships between entities are established using primary and foreign keys, which help maintain referential integrity. This structured approach allows the system to handle multiple users and bookings simultaneously while ensuring accuracy and reliability.

A key component of this project is the implementation of a stored procedure for availability check. This stored procedure is designed to determine whether rooms are available for a given date range by analyzing existing bookings. By embedding this logic within the database, the system reduces the need for complex application-side processing and improves performance. Stored procedures also enhance security and reusability, as they can be executed multiple times with different inputs without modifying the underlying code.

The system is developed using SQL-based database tools, where queries and procedures are used to manipulate and retrieve data efficiently. The availability check procedure ensures that double bookings are avoided and provides real-time responses to user requests. This feature is especially important in real-world hotel systems where accurate and quick decision-making is essential for customer satisfaction.

Overall, this project demonstrates the practical application of DBMS concepts such as schema design, normalization, and procedural SQL. The proposed solution can be further extended by integrating it with a user interface or web application, making it a complete hotel management system suitable for real-world deployment.

2 Literature Review

The development of hotel booking systems has been widely studied as part of database management and information systems research. Early systems relied on manual record-keeping or simple file-based systems, which were inefficient and prone to data redundancy and inconsistencies. With the advancement of Database Management Systems (DBMS), relational databases became the standard approach for managing structured data. Researchers have emphasized the importance of normalization, entity relationships, and integrity constraints to ensure efficient data storage and retrieval in reservation systems.

Several studies highlight the role of relational schema design in improving the performance of hotel management systems. Properly designed schemas reduce redundancy and support efficient querying of large datasets. Entities such as customers, rooms, and bookings are typically interconnected using primary and foreign keys, ensuring referential integrity. Previous works have also demonstrated how indexing and query optimization techniques can significantly enhance the speed of booking and search operations, which is critical in real-time applications.

Recent advancements focus on integrating intelligent and automated features into booking systems, such as dynamic pricing, recommendation systems, and real-time availability tracking. Although this project primarily focuses on schema design and availability checking, it aligns with these advancements by implementing an efficient stored procedure for real-time decision-making. Overall, the literature supports the idea that a well-designed database schema combined with procedural SQL can significantly improve the efficiency, reliability, and scalability of hotel booking systems.

3 Objectives

- **To Design an Efficient Hotel Booking Database** Schema

The primary objective of this project is to create a well-structured relational database schema for managing hotel operations. This includes designing tables for customers, rooms, bookings, and payments, along with defining relationships using primary and foreign keys. The goal is to ensure data integrity, reduce redundancy through normalization, and enable efficient storage and retrieval of information.

- **To Implement a Stored Procedure for Room Availability Check**

Another key objective is to develop a stored procedure that can automatically check room availability based on user-provided inputs such as check-in and check-out dates. This procedure helps in avoiding double bookings, ensures accurate results, and improves system performance by executing logic directly within the database rather than at the application level.

- **To Enhance Data Consistency and Query Performance**

The project also aims to improve overall system efficiency by optimizing queries and maintaining consistent data. By using structured schema design, constraints, and stored procedures, the system ensures reliable operations even with multiple users accessing the database simultaneously. This objective focuses on achieving faster query execution, minimizing errors, and supporting real-time booking management.

4 Methodology

4.1 Architectural Diagram

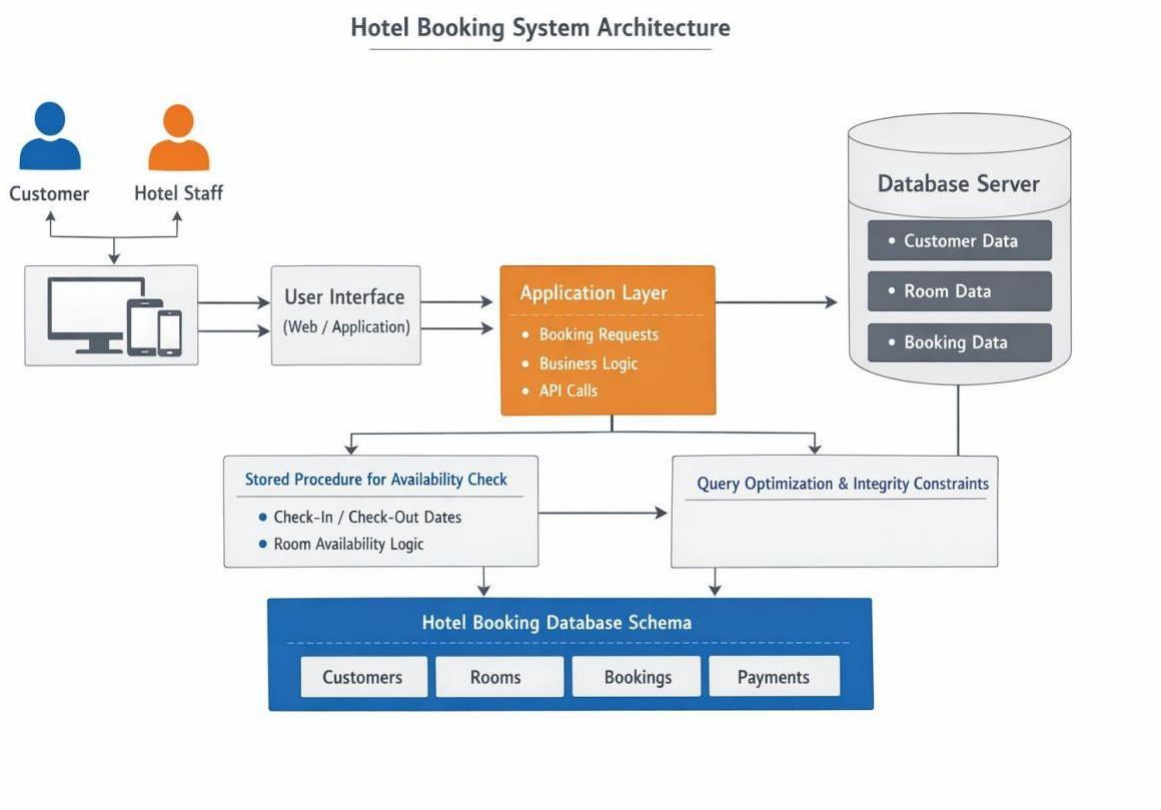

Figure 1: Architectural Diagram

The diagram shows how a Hotel Booking System works in a simple layered structure. There are two main users: customers and hotel staff, who interact with the system through a user interface such as a website or application. Their requests, like booking a room or checking details, are sent to the application layer, which processes the logic and communicates with the database server where all the data (customers, rooms, bookings) is stored.

An important part of the system is the stored procedure used to check room availability based on selected dates, which helps prevent double bookings. The database schema at the bottom organizes all the information into tables like customers, rooms, bookings, and payments. Overall, the system ensures efficient data handling, quick responses, and reliable booking management.

4.2 Hardware and Software Requirements

Hardware:

- Intel/AMD Processor
- Minimum 4GB RAM
- 500GB Hard Disk or SSD
- Standard Display (Monitor)
- Input Devices (Keyboard, Mouse)

Software:

- SQL Database (MySQL / Oracle / PostgreSQL)
- SQL Client Tool (MySQL Workbench / SQL Developer)
- Any IDE (VS Code / Eclipse / IntelliJ)
- Operating System (Windows / Linux)

4.3 Working Principle

The working principle of the Hotel Booking Database System is based on storing and managing data in a structured relational database. The system maintains tables such as customers, rooms, bookings, and payments to organize all hotel-related information efficiently.

When a customer registers, their details are stored in the Customers table. Room details like type and price are stored in the Rooms table. When a booking request is made, the system records it in the Bookings table along with check-in and check-out dates.

Before confirming a booking, a stored procedure is executed to check room availability by comparing the requested dates with existing bookings. This prevents overlapping bookings and ensures accuracy. Once confirmed, payment details are stored in the Payments table.
Using SQL queries, the system retrieves and updates data efficiently, helping manage reservations, track records, and support decision-making.

EXAMPLE TABLES

CUSTOMER TABLE

CREATE TABLE Customers (Customer_id INT PRIMARY KEY, Customer_name VARCHAR(50), Phone VARCHAR(15), City VARCHAR(50));

Customer_ID	Customer_Name	Phone	City
C101	Rahul Sharma	9876543210	Hyderabd
C102	Anjali Verma	9876543222	Bangalore

ROOM TABLE

CREATE TABLE Rooms (room_id INT PRIMARY KEY, room_type VARCHAR(30), price DECIMAL(10,2), status VARCHAR(20));

Room_ID	Room_Type	Price	Status
101	Deluxe	3000	Available
102	Standard	2000	Available
103	Suite	5000	Booked

BOOKING TABLE

CREATE TABLE Bookings (Booking_id INT PRIMARY KEY,Customer_id INT,Room_id INT,Check_in DATE, Check_out DATE,FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),FOREIGN KEY (room_id) REFERENCES Rooms(room_id));

Booking_ID	Customer_ID	Room_ID	Check_In	Check_Out
B201	C101	101	01-04-2026	05-04-2026
B202	C102	103	02-04-2026	06-04-2026

PAYMENT TABLE

CREATE TABLE Payments (Payment_id INT PRIMARY KEY, Booking_id INT,Amount DECIMAL(10,2),Payment_date DATE,FOREIGN KEY (booking_id) REFERENCES Bookings(booking_id));

Payment_ID	Booking_ID	Amount	Payment_Date
P301	B201	12000	01-04-2026
P302	B202	20000	02-04-2026

4.4 Software Implementation

```sql
-- 1. Create Database
CREATE DATABASE HotelBookingDB;
USE HotelBookingDB;

-- 2. Customers Table
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    phone VARCHAR(15),
    city VARCHAR(50)
);

-- 3. Rooms Table
CREATE TABLE Rooms (
    room_id INT PRIMARY KEY,
    room_type VARCHAR(30),
    price DECIMAL(10,2),
    status VARCHAR(20)
);

-- 4. Bookings Table
CREATE TABLE Bookings (
    booking_id INT PRIMARY KEY,
    customer_id INT,
    room_id INT,
    check_in DATE,
    check_out DATE,
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
    FOREIGN KEY (room_id) REFERENCES Rooms(room_id)
);

```

```sql

-- 5. Payments Table
CREATE TABLE Payments (
    payment_id INT PRIMARY KEY,
    booking_id INT,
    amount DECIMAL(10,2),
    payment_date DATE,
    FOREIGN KEY (booking_id) REFERENCES Bookings(booking_id)
);

-- 6. Insert Sample Data

-- Customers
INSERT INTO Customers VALUES
(101, 'Rahul Sharma', '9876543210', 'Hyderabad'),
(102, 'Anjali Verma', '9876543222', 'Bangalore');

-- Rooms
INSERT INTO Rooms VALUES
(101, 'Deluxe', 3000, 'Available'),
(102, 'Standard', 2000, 'Available'),
(103, 'Suite', 5000, 'Available');

-- Bookings
INSERT INTO Bookings VALUES
(201, 101, 101, '2026-04-01', '2026-04-05'),
(202, 102, 103, '2026-04-02', '2026-04-06');

```

```sql

-- Payments
INSERT INTO Payments VALUES
(301, 201, 12000, '2026-04-01'),
(302, 202, 20000, '2026-04-02');

-- 7. Stored Procedure for Availability Check
DELIMITER //

CREATE PROCEDURE CheckAvailability(
    IN p_check_in DATE,
    IN p_check_out DATE
)
BEGIN
    SELECT room_id, room_type
    FROM Rooms
    WHERE room_id NOT IN (
        SELECT room_id
        FROM Bookings
        WHERE (check_in < p_check_out AND check_out > p_check_in)
    );
END //

DELIMITER ;

-- 8. Execute Stored Procedure
CALL CheckAvailability('2026-04-03', '2026-04-05');

```

```sql

-- 9. Additional Queries

-- View All Bookings
SELECT * FROM Bookings;

-- View Available Rooms
SELECT * FROM Rooms WHERE status = 'Available';

-- Total Revenue
SELECT SUM(amount) AS Total_Revenue FROM Payments;
```

Run

5 Results and Discussion

Availability Check Result

```
CALL CheckAvailability('2026-04-03', '2026-04-05");
```

room_id	room_type
102	Standard

All Bookings

```
SELECT * FROM Bookings;
```

booking_id	customer_id	room_id	check_in	check_out
201	101	101	2026-04-01	2026-04-05
202	102	103	2026-04-02	2026-04-06

Available Rooms

```
SELECT * FROM Rooms WHERE status = 'Available';
```

room_id	room_type	price	status
101	Deluxe	3000	Available
102	Standard	2000	Available
103	Suite	5000	Available

Total Revenue

```
SELECT SUM(amount) AS Total_Revenue FROM Payments;
```

Total_Revenue
32000

Summary of Hotel Booking System Output

Room Availability:

Only Room 102 (Standard) is available between 2026-04-03 to 2026-04-05 because other rooms are already booked.

Bookings Data:

There are 2 bookings:

Customer 101 booked Room 101

Customer 102 booked Room 103

Available Rooms:

All rooms are marked as Available in the system, but actual availability depends on booking dates.

Total Revenue:

The hotel earned a total of ₹32,000 from all bookings.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Apply database concepts like SQL, tables, and relationships to develop the hote booking system.
PO2	Problem Analysis	Identify and analyze booking, availability, and payment management problems in hotel systems.
PO3	Design/Development of Solutions	Design an efficient database structure to handle customers, rooms, bookings, and payments.
PO4	Conduct Investigations of Complex Problems	Test queries and procedures to ensure correct outputs like availability and revenue.
PO5	Engineering Tool Usage	Use tools like MySQL and database software to implement and manage the system.

6.2 Sustainable Development Goals (SDGs) Mapping

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	Goal	Relevance
SDG 8	Decent Work and Economic Growth	This system improves hotel operations by automating booking and management processes. It helps increase revenue and supports growth in the hospitality sector.
SDG 9	Industry, Innovation and Infrastructure	The use of database systems promotes innovation and digital transformation in hotel management. It enhances efficiency and strengthens technological infrastructure.
SDG 12	Responsible Consumption and Production	The system ensures optimal use of rooms and reduces unnecessary It supports sustainable practices by improving resource management.

7 Conclusion

The Hotel Booking Database System successfully demonstrates how structured data management can improve hotel operations. By using relational tables for customers, rooms, bookings, and payments, the system ensures efficient storage, retrieval, and processing of information. Features like availability checking and revenue calculation help automate key tasks, reduce manual effort, and minimize errors. The implementation shows how SQL queries and procedures can be effectively used to handle real-world scenarios in the hospitality domain.

Overall, the system provides a reliable and scalable solution for managing hotel bookings. It enhances operational efficiency, supports better decision-making, and ensures proper utilization of resources. The project also highlights the importance of database design and modern tools in developing practical applications, making it useful for both academic learning and real-world implementation.

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